Paper 1 Section B

Marks

1. (a) $(1.5 \times 1000\ \text{kg}) \times 4200\ \text{J kg}^{-1}\ ^\circ\text{C}^{-1} \times (80 - 60)\ ^\circ\text{C} \times (1-15\%)$ 1M+1M
$= 1.07 \times 10^8\ \text{J}$ 1A 3

(b) $1.07 \times 10^8\ \text{J} \div (4.5\ \text{kW}) \div 3600\ \text{s}$ 1M
$= 6.61$ (hours) 1A 2

(c) Rate of heat transfer drops as the water temperature drops / the room temperature increases / temperature difference drops. 1A 1

2. (a) Stop heating and — 1A
stir the water well. — 1A 2

(b) (i) $\frac{L-64}{80-64}=\frac{65-20}{92-20}$ — 1M

$L-64=10$

$L=74$ mm — 1A 2

(ii) Let x °C be the absolute zero.

$\frac{20-x}{92-20}=\frac{64-0}{80-64}$ — 1M

$20-x=288$

$x=-268$ °C — 1A 2

3. (a) (i) $P = F v$ 1M

$= 8000\ \text{N} \times 2\ \text{m s}^{-1}$

$= 16\ \text{kW}$ 1A 2

(ii) $P_{\text{loss}} = 20\ \text{kW} - 16\ \text{kW} = 4\ \text{kW}$ 1A 1

(b) (i) $P = 4\ \text{kW} + (8000 - 7000\ \text{N}) \times 2\ \text{m s}^{-1}$ 1M

$= 6000\ \text{W} = 6\ \text{kW}$ 1A 2

(ii) Output power required from motor is smaller. 1A

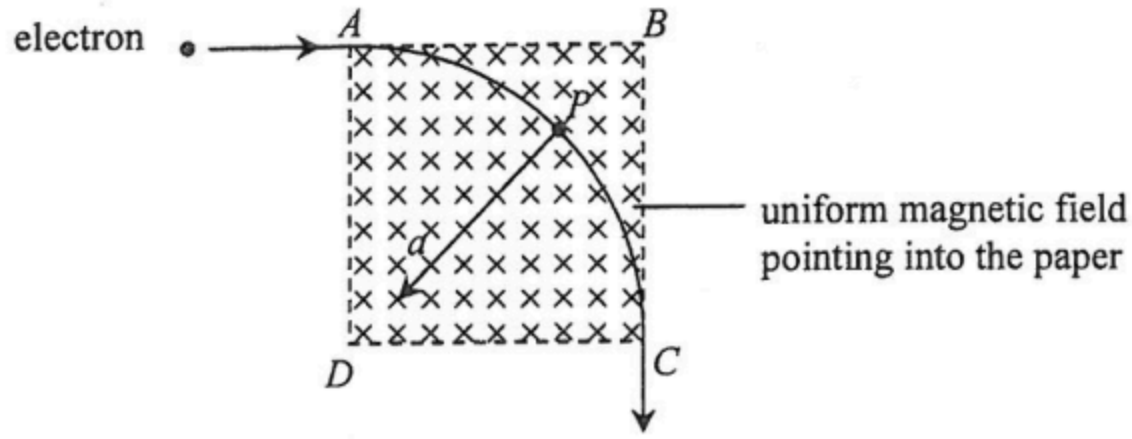
Or Force exerted by motor is smaller. 1

(iii) No, the lift system could not work / the lift will fall 1A

as slipping would occur / the cable cannot be fixed on the drum / the drum cannot exert a force on the cable. 1A 2

4. (a) (i) $F = qvB = (1.60\times10^{-19}\text{ C})(1.2\times10^{7}\text{ m s}^{-1})(0.01\text{ T})$ 1M
$= 1.92\times10^{-14}\text{ N}$ 1A 2

(ii)



1A 1

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Marks

4. (b) F is perpendicular to the electron's displacement / velocity, 1A
Or electron only changes direction while speed remains unchanged,
no work is done, therefore k.e. remains unchanged. 1A 2

(c)
$$F = qvB = \frac{mv^2}{r}$$
$$\frac{v}{r} = \frac{qB}{m} = \text{constant}$$
} 1M

1A 2

v should be halved, i.e. $0.6\times10^{7}\text{ m s}^{-1}$

5. Stick the paper strip onto the horizontal part BC of the track. 1A

Release the toy from a certain height h from the bench surface and measure the corresponding stopping distance d. 1A

Stopping distance should be measured from the beginning of the horizontal part BC or on the paper strip.

Release the toy from different heights and measure the corresponding stopping distances. 1A

Plot a graph of d against h, 1A

a straight line passing through the origin should be obtained. 1A

or

since $mgh = Fd$

so $\frac{d}{h} = \text{constant} \ / \ d \propto h$

5

6.	(a)	(i)	Accelerates at g before the elastic cord stretches / at the beginning.	1A		
			Acceleration decreases as the cord stretches.	1A		
			Decelerates until momentarily at rest (after the tension in the cord is greater than mg).	1A		3
		(ii)	Gravitational potential energy changed to kinetic energy and	1A		
			(then) elastic potential energy in elastic cord.	1A		2
	(b)		Elastic cord lengthens the stopping time,	1A		
			hence reduces the (net) force acting on the player.	1A		2
	(c)		Contact area is larger,	1A		
			hence pressure is smaller during the fall and the structure is less likely to break / detach.	1A		2

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Marks

7. (a) $c = f\lambda \Rightarrow 3 \times 10^8 \text{ m s}^{-1} = f(0.02 \text{ m})$ 1M

$\therefore \quad f = 1.5 \times 10^{10} \text{ Hz or } 15000 \text{ MHz}$ 1A 2

(b) (i) Path difference of the diffracted waves from slits A and B to probe varies along XY. 1A

Constructive and destructive interference occur alternately to give maxima and minima. 1A 2

(ii) $BP - AP = 1\frac{1}{2}\lambda$ 1M

$BP - AP = 3 \text{ cm} = 0.03 \text{ m}$

$\therefore BP = 1.24 + 0.03 = 1.27 \text{ m}$ 1A 2

(iii) Path difference along $XY < AB$ } 1M

$AB = 3 \times 2 \text{ cm} = 3\lambda$

$\therefore$ path difference allowed = $0\lambda, 1\lambda, 2\lambda$. }

Maximum number of maxima = 3 1A 2

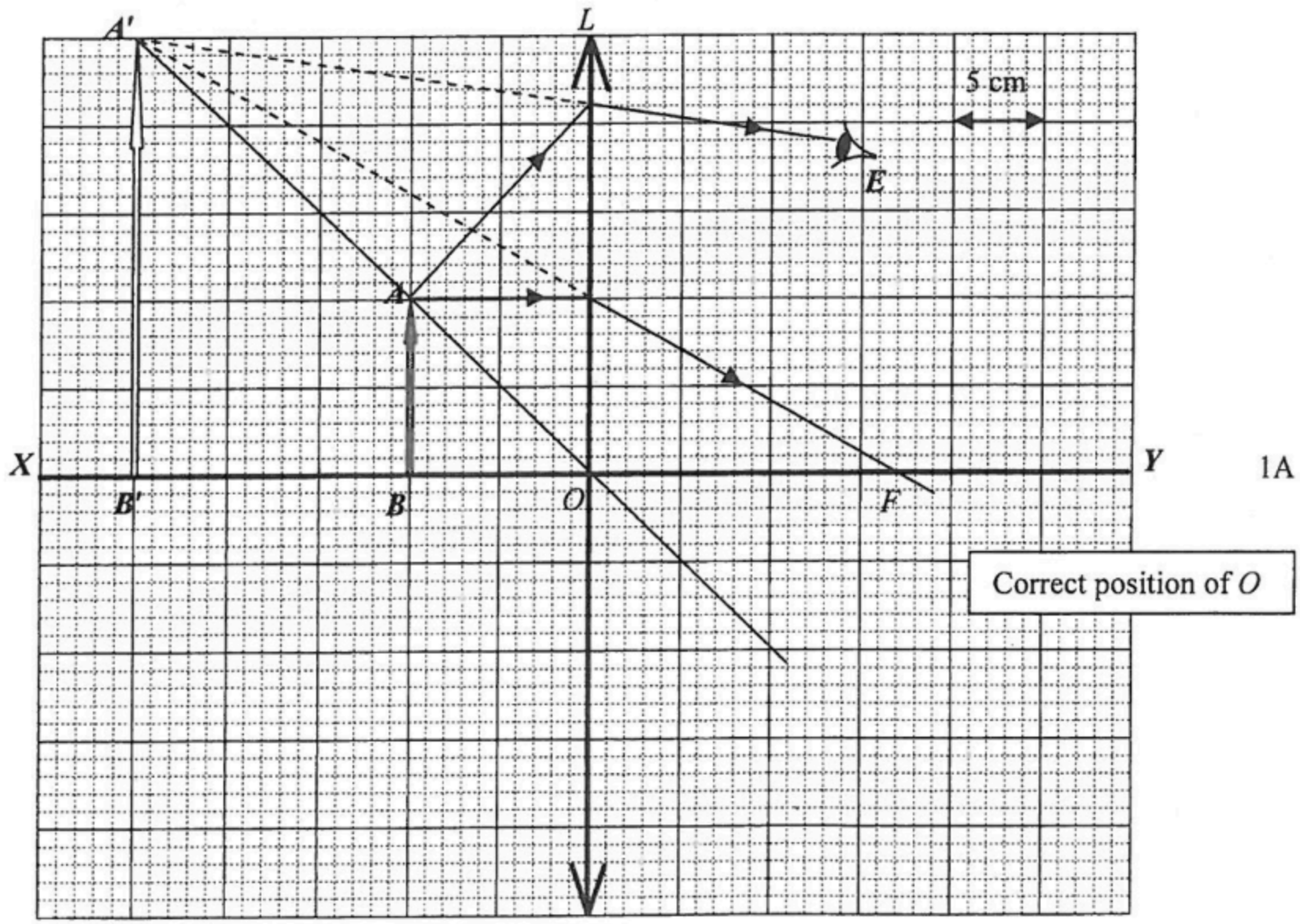
(c) Radio waves with lower frequencies (will have longer wavelengths and hence) have greater diffraction effect. 1A

Radio waves by-pass small obstacles / not to be reflected from small obstacles. 1A 2

8. (a) (i) Virtual 1A 1

(ii) Convex. 1A
Only convex lens can form magnified (virtual, erect) images. 1A 2

(b) (i)



1A 1

Correct position of *O*

(ii) Correct light ray to locate *F*. 1M
Focal length *f* = 17 cm (16.0 to 17.5 cm) 1A 2

(c) Correct ray from *A'* or lens to *E*. 1A
All correct. 1A 2

(d) Magnifying glass / glasses for long-sighted eyes / simple microscope 1A 1

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Marks

9. (a) $$k = \frac{\ln 2}{5730 \times 3.16 \times 10^{7}} = 3.83 \times 10^{-12}\,(\text{s}^{-1})$$ 1A

Activity $A = kN$

$$N = \frac{A}{k} = \frac{0.2}{3.83 \times 10^{-12}}$$ 1M

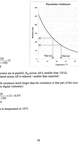
$$= 5.22 \times 10^{10}$$ 1A 3

(b) No. of ^{14}C nuclei: $N_0 = 1 \times 10^{23} \times (1.3 \times 10^{-12}) = 1.3 \times 10^{11}$ 1A 1

(c) $$kt = \ln \frac{N_0}{N}$$

$$(3.83 \times 10^{-12})\ t = \ln \frac{1.3 \times 10^{11}}{5.2 \times 10^{10}}$$ 1M

$t = 2.4 \times 10^{11}$ s or 7571 (years) (accept 7500 to 7600 (years)) 1A 2



(c) $E_{photon} = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{1.1 \times 10^{-6}} = 1.8 \times 10^{-19}$ J 1M 1A

(d) As E and 1/λ (I) increase up to a plateau E_g across cell is smaller than 1/λ (I). 1A
Above plateau E_g remains constant and E_{photon} increases. 1A

(e) Photons with energy less than E_g do not have sufficient energy to excite an electron from the valence band to the conduction band. 1A

(f) $P_{max} = \frac{V_{max} \times I_{max}}{A} = \frac{0.5 \times 0.03}{0.0001} = 150$ W m^{-2} 1M 1A

corresponds to temperature of 18°C. 1A

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Marks

11 (a) (i) Current circuit (i.e. wire/charge detection X and current source). 1A
As the temperature drops, the thermistor resistance increases. 1A
Thus the resistance increases to a value such that $I_{TH} < I_{REF}$ X is also, the electronic switch is set and closes at the heating device. 1A

11 (a) (ii) $I_{TH} = I_{REF}$ 1M
$\frac{V_{TH}}{R_{TH}} = \frac{V_{REF}}{R_{REF}}$ 1M
$R_{TH} = \frac{V_{TH} \times R_{REF}}{V_{REF}} = \frac{2.5 \times 10^{3}}{5} = 500\ \Omega$ 1A

(b) (i) Potential set $= \frac{R_2}{R_1 + R_2} \times V_{out} = \frac{10^{3}}{10^{3} + 10^{3}} \times 5 = 2.5$ V 1M
$= 2.5$ V 1A

(ii) Separation of between 1A

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Paper 2

Section A - Astronomy and Space Science

1. B [1A]	2. E [1A]	3. A [1A]	4. E [1A]
5. B [1A]	6. B [1A]	7. C [1A]	8. E [1A]

Marks

1. (a) Satellite will be directly above a certain location on the equator of the Earth, 1A
and will orbit in the same direction as the Earth's rotation. 1A
Any satellite orbit maintaining a matching angular from the Earth's rotation about the axis of rotation. 1A
Accept: Satellite remains in the same position relative to the Earth's surface.

(b) $\frac{GMm}{r^2} = \frac{mv^2}{r}$ 1M
$v = \sqrt{\frac{GM}{r}}$
$= \sqrt{\frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{6.8 \times 10^{6}}}$
$= 7753$ m s^{-1} 1A 1A

(c) Total energy $= \frac{1}{2}mv^2 - \frac{GMm}{r}$ 1M
$= \frac{1}{2} \times 1000 \times 7753^2 - \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1000}{6.8 \times 10^{6}}$
$= -2.4 \times 10^{10}$ J 1M 1A

(d) $\Delta E = \frac{GMm}{r_1} - \frac{GMm}{r_2} = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$ 1M
$= 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1000 \left(\frac{1}{6.8 \times 10^{6}} - \frac{1}{6.9 \times 10^{6}}\right)$
$= 1.4 \times 10^{8}$ J 1A

(e) Angle is small but the aligned value of $\frac{r^3}{T^2}$ 1M
$\frac{r^3}{T^2} = \frac{(6.8 \times 10^{6})^3}{(5.8 \times 10^{3})^2}$
$= 9.4 \times 10^{12}$
Time from A to B $= \frac{\text{Area of ellipse}}{\text{Area of circle}} \times \text{Time for circle} = \frac{\frac{1}{2} \times \pi \times 6.8 \times 10^{6} \times 6.5 \times 10^{6}}{\pi \times (6.8 \times 10^{6})^2} \times 5.8 \times 10^{3}$
$= 2.8 \times 10^{3}$ s 1M 1A

(f) (i) $\frac{\Delta E}{\Delta t} = \frac{\text{Energy released by sun in one second}}{\text{Area of sphere of radius 1 AU}}$ 1M
$= \frac{3.8 \times 10^{26}}{4\pi \times (1.5 \times 10^{11})^2}$
$= 1350$ W m^{-2} 1A

(ii) If a single photon has sufficient energy to knock out an electron, the electron gains enough KE to escape the metal surface. 1A
Increasing the intensity of the radiation increases the number of electrons emitted but not the energy of each one. 1A
Hence, it is found that the work function of the metal is 1.4 eV. 1A

(g) (i) $\Delta E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{656 \times 10^{-9}} = 3.03 \times 10^{-19}$ J 1M
$= 1.9$ eV (answered not needed) 1A

$E_{KE} = 1.9^{eV} - 1.4 \text{ eV} = 0.5^{eV}$ 1A
$= 0.5 \times 1.6 \times 10^{-19}$ J 1A

(ii)

1 reference only 1A

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Section B - Atomic World

9. E [1A]	10. C [1A]	11. C [1A]	12. A [1A]
13. A [1A]	14. D [1A]	15. C [1A]	16. D [1A]

Marks

2. (a) $\frac{\Delta N(t)}{\Delta t} = \lambda N = \ln 2 \times \frac{N}{t_{1/2}}$ 1A
$= \ln 2 \times 6.0 \times 10^{23} \div 2.1 \times 10^{5}$ 1A

(b) Only those molecules / free electrons at the surface can have the maximum kinetic energy. 1A
(i) The total fraction of a metal is only the minimum energy required to eject an electron. 1A
(ii) Some energy is lost as the electron moves through the metal. 1A
(iii) Less energy is required to eject an electron from the excited state and requires more energy to break the bonds. 1A
(iv) Some electrons are not at the surface of metal so don't have maximum kinetic energy. 1

(c) Energy absorbed by an atom is $\frac{hc}{\lambda}$ 1M
$\Delta E = h \times c \div \lambda = 6.63 \times 10^{-34} \times 3.0 \times 10^{8} \div 1.5 \times 10^{-10}$ 1A
$= 1.3 \times 10^{-15}$ J 1A

(d) (i) If a single photon has sufficient energy to knock out an electron, the electron gains enough KE to escape the metal surface. 1A
Increasing the intensity of the radiation increases the number of electrons emitted but not the energy of each one. 1A
Hence, it is found that the work function of the metal is 1.4 eV. 1A

(ii) $\Delta E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{656 \times 10^{-9}} = 3.03 \times 10^{-19}$ J 1M
$= 1.9$ eV (answered not needed) 1A

$E_{KE} = 1.9^{eV} - 1.4 \text{ eV} = 0.5^{eV}$ 1A
$= 0.5 \times 1.6 \times 10^{-19}$ J 1A

(iii)

1 reference only 1A

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Section C - Energy and Use of Energy

17. A [1A]	18. B [1A]	19. C [1A]	20. E [1A]
21. E [1A]	22. A [1A]	23. C [1A]	24. D [1A]

Marks

3. (a) $\frac{\Delta N(t)}{\Delta t} = \lambda N = \frac{\ln 2}{t_{1/2}} \times N$ 1M
$= \ln 2 \div 5730 \times 6.0 \times 10^{23}$ 1A
$= 7.2 \times 10^{19}$ 1A

(b) Rough archer should be rated much that 1A
Ambulance becomes efficient to reduce space. 1A

(c) (i) $\Delta E = \Delta M \times (3 \times 10^{8})^2 = 4.5 \times 10^{16} \times 9 \times 10^{16}$ 1M
$= 4.1 \times 10^{33}$ J (accept 4.0×10^{33} or 10^{33}) 1A

(ii) $\frac{4.1 \times 10^{33}}{3.8 \times 10^{26}} = 1.1 \times 10^{7}$ s = 0.35 Y 1M
$= 4.1 \times 10^{33}$ J (accept 4.0×10^{33} or 10^{33}) 1A

(d) Windows scale from a ceiling. 1A
(i) Thermal radiation. 1A
(ii) Reduces the Stefan-Boltzmann law. 1A
(iii) Replaces air conditioner with lighter-cooling capacity / LED 1

4. A particle in a diffusion fluid cloud of oil droplets. It takes from a 1A Part II and then is in full from 1A
F & E & G/T 1A / N / Part I, 1A Magnetic air resistance 1A

(a) $F = \frac{1}{2}\rho v^2 C_D A$ 1M
$= \frac{1}{2} \times 1.2 \times 30^2 \times 0.4 \times 0.5$ 1A
$= 108$ N 1A

(b) $P = \frac{W}{t} = \frac{F \times s}{t} = F \times v$ 1M
$= 108 \times 30$ 1A
$= 3.2$ kW 1A

(c) $\frac{1}{2}\rho v^2 C_D A = \frac{1}{2} \times 1.2 \times 60^2 \times 0.4 \times 0.5 = 432$ N 1M
$P = F \times v = 432 \times 60 = 25.9$ kW 1A

(d) Reduces fuel consumption 1A
(i) Reduces emissions with higher-cruising capacity / LED 1

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Section D - Medical Physics

25. A [1A]	26. E [1A]	27. C [1A]	28. A [1A]
29. A [1A]	30. B [1A]	31. C [1A]	32. D [1A]

Marks

4. (a) (i) 2.25 cm 1A

(ii) $\frac{v_1}{\sin\theta_1} = \frac{v_2}{\sin\theta_2}$ (or $n_1 v_1 = n_2 v_2$) 1M
$\frac{3.0 \times 10^{8}}{\sin 30°} = \frac{v_2}{\sin 20°}$
$v = 2.05 \times 10^{8}$ (accept 2.0×10^{8}) 1A

(iii) Medium 2 / lower density 1A

(b) (i) Intensity of X-ray is attenuated / absorbed when they pass through a medium. 1A
Bone is more dense than soft tissue / fat / muscle, so it absorbs more X-rays. Therefore the film appears lighter under these / darker under soft tissue. 1A

(ii) The X-ray tube and detectors rotate round the patient to take multiple X-ray projections / 1A
The projections are used to reconstruct isotopes / make fluid projection / isotherm to form cross-sectional images of the patient's body. 1A

(iii) radiation exposure or dosage is much higher for CT scan (by 8 with 4x) 0.01 safer for X-ray 1A
CT scanner is usually accessible at X-ray imaging. 1A

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Candidates' Performance

Paper 1

Paper 1 consists of two sections, multiple-choice questions in Section A and conventional questions in Section B. Candidates are required to answer all questions.

Section A: multiple-choice questions

Section A consisted of 32 multiple-choice questions and the mean score was 70. Candidates' performance in the following items scored a mean of four marks/ten:

5.

A female on a rough horizontal surface is moving to the left with constant speed under two horizontal forces of 10 N and 15 N, as shown in the figure. Given that $g = 10$ m s^{-2} is constant, what is the net force exerted by the surface on the person?

A. 5 N (25%)
B. 10 N (25%)
C. 15 N (25%)
D. 25 N (25%)

Only one quarter of the candidates identified the correct answer as Option A, which suggests that a majority of candidates did not realise that friction applies as long as the block is still moving.

6.

A particle in a frictionless floor out of oil droplets. It takes from a 1A Part II and then is in full from F & E & G/T 1A / N / Part I, 1A Magnetic air resistance

(A) 0 N (25%)
(B) 1 N (25%)
(C) 2 N (25%)
(D) 3 N (25%)

It seems that candidates were not competent in using the formula $F = \frac{1}{2}\rho v^2 C_D A$ to tackle this problem.

15

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A particle is projected into the air at time $t = 0$ and it performs a periodic motion before landing on the ground as shown. Which graph represents the variation of the kinetic energy (KE) of the particle with time before it lands? Neglect air resistance.

More than half of the candidates did not realise that the graph should also have energy when it flies horizontally at the highest point.

16.

A point light source of 2 m the principal axis of a thin convex lens refers to ray of light. The ray passes through the lens and emerges parallel to the principal axis. The distance of the object from the lens is 30 cm and the focal length of the lens is 15 cm. Which of the following statements is true?

(A) The focal length of the lens is shorter than 20 cm.
(B) The lens is a diverging lens because it produces a virtual image.
(C) An object placed at 1 m produces a real image at 0.3.

A. (C) only (25%)
B. (A) and (B) only (25%)
C. (A) and (C) only (25%)
D. (A) and (C) only (25%)

About 60% of the candidates did not demonstrate a full understanding of the concept of reversibility of light.

17.

A straight wire carrying current I pointing into the paper is placed in a magnetic field between pole pieces A, B, C and D. Which pole pieces should be magnetised as north pole in order to produce the largest force on the wire? Ignore the effect of the Earth's magnetic field.

	polarity of F	direction of magnetic force
A.	A only	(25%)
B.	A and C	(25%)
C.	B and D	(25%)
D.	C only	(25%)

More than half of the candidates identified the correct polarity although some of them made mistakes in the direction of magnetic force.

18.

A square coil XY is placed in a pair of smooth inclined conducting rails which are bowed in a magnetic field that is oscillating between A, B, C and D. Which diagram shows how the induced current in the coil varies with time? Take clockwise current as positive. Which diagram shows how the induced current in the coil varies with time? Take clockwise current as positive.

More than 30% of the candidates did not realise that the magnetic flux Φ_B must be perpendicular to the magnetic field.

19.

A metal rod AB is released about AB in a clockwise direction in the plane of the paper with a uniform magnetic field pointing into the paper. Which statement is correct?

A. An induced current flows in the rod from A to B. (25%)
B. An induced current flows in the rod from B to A. (25%)
C. It is not correct to say that an induced current flows in the rod. (25%)
D. No induced current flows in the rod. (25%)

More than 30% of the candidates wrongly thought that an induced current could flow in an incomplete circuit.

20.

As made all the circuit shown, the coil has constant cut it P and negligible internal resistance. When the switch is closed, the current increases. Which of the arrangements is the right, how would the brightness of each bulb change?

	bulb L_1	bulb L_2
A.	becomes dimmer	remains unchanged (25%)
B.	becomes brighter	becomes brighter (25%)
C.	becomes brighter	becomes dimmer (25%)
D.	becomes dimmer	remains unchanged (25%)

More than 40% of the candidates failed to realise that the brightness of L_2 would not be affected by the change in resistance of L_1 in the parallel branch.

21.

In the above circuit, the coil has negligible internal resistance. When switch S is closed, both bulbs are not lit. Which statement is correct? Assume that the bulbs have the same value. If only one bulb has been damaged, which of the following is possible?

A. Bulb A has been damaged and is open-circuited. (25%)
B. Bulb B has been damaged and is open-circuited. (25%)
C. Bulb A is short-circuited. (25%)
D. Bulb B is short-circuited. (25%)

It seems that more than half of the candidates did not fully understand the reasoning of short circuit and open circuit.